

ME 7751
Homework 4

See homework submission guidelines on Canvas.

Problem 1

Consider a two-dimensional steady-state flow of a viscous, incompressible Newtonian fluid entering a straight channel of height H and length L . The fluid enters the channel with a uniform inlet velocity U . The flow is characterized by a Reynolds number $Re = \rho UH/\mu$, where ρ and μ are the fluid density and viscosity, respectively.

1. Solve the above problem using the artificial compressibility method. Describe the essential steps of the method in physical terms, explaining why it can be used to model an incompressible fluid.
2. Describe the dimensionless governing equations and boundary conditions. Discretize the governing equations to achieve second order spatial accuracy. Employ a collocated grid arrangement for the domain discretization, and use ghost nodes to enforce boundary conditions.
3. Compute the steady state solutions for $Re = 20$. Set $H = 1$ and select the length of the domain L sufficiently large that the flow is fully developed at least the distance H from the outlet. Use $N = 20$ grid points across the channel height. Explain how to determine when a flow reaches the fully developed regime. Compare the fully developed flow velocity profile with the analytical solution for channel flow.
4. Plot velocity profiles at different locations along the channel. Plot pressure distribution along the channel centerline. Explain the result.
5. Discuss the stability criteria of the method and explain how the time step is selected.
6. Numerically compute the spatial convergence rate of your model by performing simulations with progressively increased vertical grids consisting of $N = 10, 20, 40$, and 60 nodes. Evaluate and plot on log-log plot the solution error relative the solution with the highest grid resolution. Detail how you computed the spatial convergence rate.
7. Evaluate the entrance length L_e for Reynolds numbers ranging from 1 to 500. Use a mesh with $N = 20$ and adjust the channel length L to ensure fully developed flow at the channel outlet. Plot how L_e depends on the Reynolds number and compare your results with the empirical correlation $L_e/H = [0.631^{1.6} + (0.0442Re)^{1.6}]^{1/1.6}$.^a

^a Durst F., Ray S., Ünsal B., and Bayoumi O. A. "The Development Lengths of Laminar Pipe and Channel Flows." ASME. J. Fluids Eng. 2005; 127(6): 1154-1160.